

Intensive prolonged exposure treatment for adolescent complex posttraumatic stress disorder: a single-trial design

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Background: The current study evaluated the effectiveness and safety of intensive prolonged exposure (PE) targeting adolescent patients with complex posttraumatic stress disorder (PTSD) and comorbid disorders following multiple interpersonal trauma. **Methods:** Ten adolescents meeting full diagnostic criteria for PTSD were recruited from a specialized outpatient mental health clinic and offered a standardized intensive PE. The intensive PE consisted of three daily 90-min exposure sessions delivered on five consecutive weekdays, followed by 3 weekly 90-min booster sessions. In a single-trial design, the participants were randomly allocated to one of five baseline lengths (4–8 weeks) before starting the intensive PE. Before, during, and after intensive PE completion, self-reported PTSD symptom severity was assessed weekly as a primary outcome (a total of 21 measurements). Furthermore, clinician-administered PTSD diagnostic status and symptom severity (primary outcome), as well as self-reported comorbid symptoms (secondary outcomes), were assessed at four single time points (baseline-to-6-month follow-up). **Results:** Time-series analyses showed that self-reported PTSD symptom severity significantly declined following treatment ($p = .002$). Pre-postgroup analyses demonstrated significant reductions of clinician-administered PTSD symptom severity and self-reported comorbidity that persisted during the 3- and 6-month follow-ups (all $ps < .05$), where 80% of adolescents had reached diagnostic remission of PTSD. There was neither treatment dropout nor any adverse events. **Conclusions:** The results of this first proof of concept trial suggest that intensive PE can be effective and safe in an adolescent population with complex PTSD, although the gains achieved need to be confirmed in a randomized controlled trial. **Keywords:** Complex PTSD; treatment outcome; adolescents; prolonged exposure; intensive treatment.

Introduction

Posttraumatic stress disorder (PTSD) is a severe and very debilitating disorder. A meta-analysis evaluating 43 independent samples showed that the overall PTSD rate in trauma-exposed children and adolescents was 15.9%, with children and adolescents having been exposed to interpersonal trauma, for example sexual and physical abuse, showing the highest risk (Alisic et al., 2014).

In clinical guidelines across the world, trauma-focused therapy (TFT) is considered the first-line treatment for these children and adolescents (International Society for Traumatic Stress Studies, Foa, Keane, Friedman, & Cohen, 2009; National Institute for Health and Clinical Excellence Guidelines on PTSD, NICE, 2005). A recent meta-analysis evaluating 14 studies describing the effects of PTSD treatments in children and adolescents indeed concluded that TFT provided the best evidence of effectiveness (Gillies, Taylor, Gray, O'Brien, & D'Abrew, 2013). Proven effective TFT protocols include the trauma-focused cognitive behavioral therapy (TF-CBT) of Cohen, Mannarino, and Deblinger (2006), comprising

gradual exposure and other trauma-focused techniques (e.g. cognitive processing), and the prolonged exposure (PE) protocol by Foa, McLean, Capaldi, and Rosenfield (2013). In both protocols, treatment is delivered in weekly sessions over the course of several months.

Given that most RCTs thus far focused on the effectiveness of TFT for children or adolescents with PTSD following single trauma exposure (Gillies et al., 2013), the effectiveness of such programs for young patients having suffered multiple trauma is still largely unknown. There is also only limited research available about the effectiveness of TFT in young populations showing symptoms of complex PTSD, such as affect dysregulation, negative self-concept, and interpersonal problems, or in populations showing other comorbid symptoms (Smith et al., 2013).

In response to this research gap, it has recently been suggested that, for safety reasons, in clinical practice, a phase-based approach, in which a stabilizing phase is offered prior to starting the trauma-focused treatment phase, would be advisable for this group of complex PTSD patients (Leenarts, Diehle, Doreleijers, Jansma, & Lindauer, 2013). This recommendation was partly based on the incorporation of additional modules (e.g. affective coping

Conflict of interest statement: No conflicts declared.

techniques) in the most researched TF-CBT protocol of Cohen et al. (2006); however, the effects of those additional modules have not been clarified (Leenarts et al., 2013). Furthermore, as a stand-alone treatment, PE is underutilized in clinical practice in general and clinicians perceive barriers to the treatment in terms of dropout, adverse events, and symptom exacerbation (Van Minnen, Hendriks, & Olff, 2010).

Because it is especially children and adolescents who are exposed to interpersonal trauma that are most likely to develop PTSD, it is of paramount importance to establish TFT effectiveness in pediatric patients with complex PTSD following interpersonal traumas. Furthermore, in view of the recent suggestions of a phase-based approach and the lack of evidence supporting its benefits, it would be interesting to learn whether stand-alone TFT, that is without any stabilizing phase or additional module(s), is safe for this population.

A factor that stands in the way of an effective treatment of PTSD in adolescents is premature discontinuation. Dropout rates were as high as 33% in the studies using Cohen's TF-CBT protocol (Cohen, Mannarino, & Iyengar, 2011) and 21% for PE (Gilboa-Schechtman et al., 2010). A plausible explanation for these high dropout rates is lengthy treatment duration, which increases the likelihood of dropout. This would be in line with a meta-analysis of PTSD effect studies that showed that dropout among adult patients increased with increasing numbers of (weekly) sessions, and thus longer therapy duration (Imel, Laska, Jakupcak, & Simpson, 2013), warranting the assumption that premature termination may be prevented by treatments that offer sessions in a shorter period of time. Intensive TFT programs for adults with PTSD have recently been developed and tested, and one RCT found that intensive cognitive therapy delivered within a 1-week timeframe was equally effective in reducing PTSD symptoms as a regular weekly-based protocol (Ehlers et al., 2014). Other studies were exploratory and also showed that intensive TFT reduced PTSD symptoms (Hendriks, de Kleine, Hendriks, & van Minnen, 2016). More importantly, very few patients left the programs early and adverse events did not occur. There are two uncontrolled PTSD studies that showed an intensive TFT delivered in the course of 4 weeks to be effective in reducing self-reported and clinician-rated PTSD symptoms in children and adolescents having experienced multiple interpersonal traumas (Matulis, Resick, Rosner, & Steil, 2014; Silverstone, Greenspan, Silverstone, Sawa, & Linder, 2016). Although these initial outcomes are encouraging, the effectiveness of intensive TFT for pediatric PTSD populations needs to be further established.

The purpose of the present study accordingly was to investigate the effectiveness and safety of an intensive PE therapy for adolescents in an experimental design

that controlled for time effects. The program was specifically developed for adolescents who developed PTSD due to multiple interpersonal traumas and who show high levels of complex PTSD symptoms as well as fulfilling criteria for other comorbid disorders. We hypothesized that our intensive PE would ameliorate their PTSD symptoms, as well as depressive, general anxiety, and dissociative symptoms, and second, that it would be low in dropout and safe.

Methods

The trial design was approved by the medical ethics committee of the Radboud University Medical Centre and registered in the Dutch trial register with number NTR4301 (trialregister.nl).

Participants

The participants ($N = 10$) were regular referrals to a Dutch outpatient mental health clinic specialized in the treatment of anxiety disorders who were recruited between 2013 and 2014 (see participant flow diagram, Figure 1). Inclusion criteria were (a) age 12–18 years; (b) history of multiple interpersonal trauma; (c) meeting full DSM-IV-TR (APA, 2000) diagnostic criteria for PTSD established through the Kiddie-Schedule for Affective Disorders and Schizophrenia-Present and Lifetime Version PTSD section (K-SADS-PL-PTSD; Kaufman et al., 1997); (d) written informed adolescent and (if under 18) parental consent. Exclusion criteria were (a) suicide attempt within 8 weeks prior to study entry (i.e. suicidality without imminent threat was not an exclusion criterion); (b) inability to speak and write Dutch; (c) mental retardation ($IQ \leq 70$); (d) changes in psychoactive medication within 8 weeks prior to enrollment; (e) ongoing trauma by a caregiver who is part of the child's current primary-care system; (f) current DSM-IV PTSD diagnosis of the caregiver when involved in the child's current treatment as established with the Mini-International Neuropsychiatric Interview (M.I.N.I.; Sheehan et al., 1998) as well as a clinical cut-off score of ≥ 20 on the PTSD self-report symptom scale (PSS-SR; Foa, 1995; Foa, Riggs, Dancu, & Rothbaum, 1993).

In view of our aims, the presence of complex PTSD or other comorbid disorders was no exclusion criterion. The DSM-IV Axis I diagnoses of any comorbid disorders were assessed with the K-SADS-PL (Kaufman et al., 1997).

Treatment

Our intensive PE therapy program was largely based on Foa's PE protocol (Foa, Hembree, & Rothbaum, 2007), with the difference that, instead of weekly sessions, the intervention was delivered in five successive weekdays (of the same week) and comprised a total of 15 PE sessions. Each of the five treatment days, the adolescents received three individual, manualized 90-min sessions. The first of these daily sessions comprised prolonged imaginal exposure during which the participant was asked to recount aloud the traumatic memory as vividly as possible, with closed eyes, in the present tense, and as detailed as possible. The second session concerned exposure by drawing in that the participants were instructed to draw the scene(s) of the trauma-memory hotspots and to include all anxiety-provoking details of this hotspot in the drawing. The third session of the day involved exposure *in vivo*, where participants were exposed to trauma-related material and situations. As an add-on to the manualized PE treatment, parents attended daily sessions involving psychoeducation and social support skill training, which we included based on

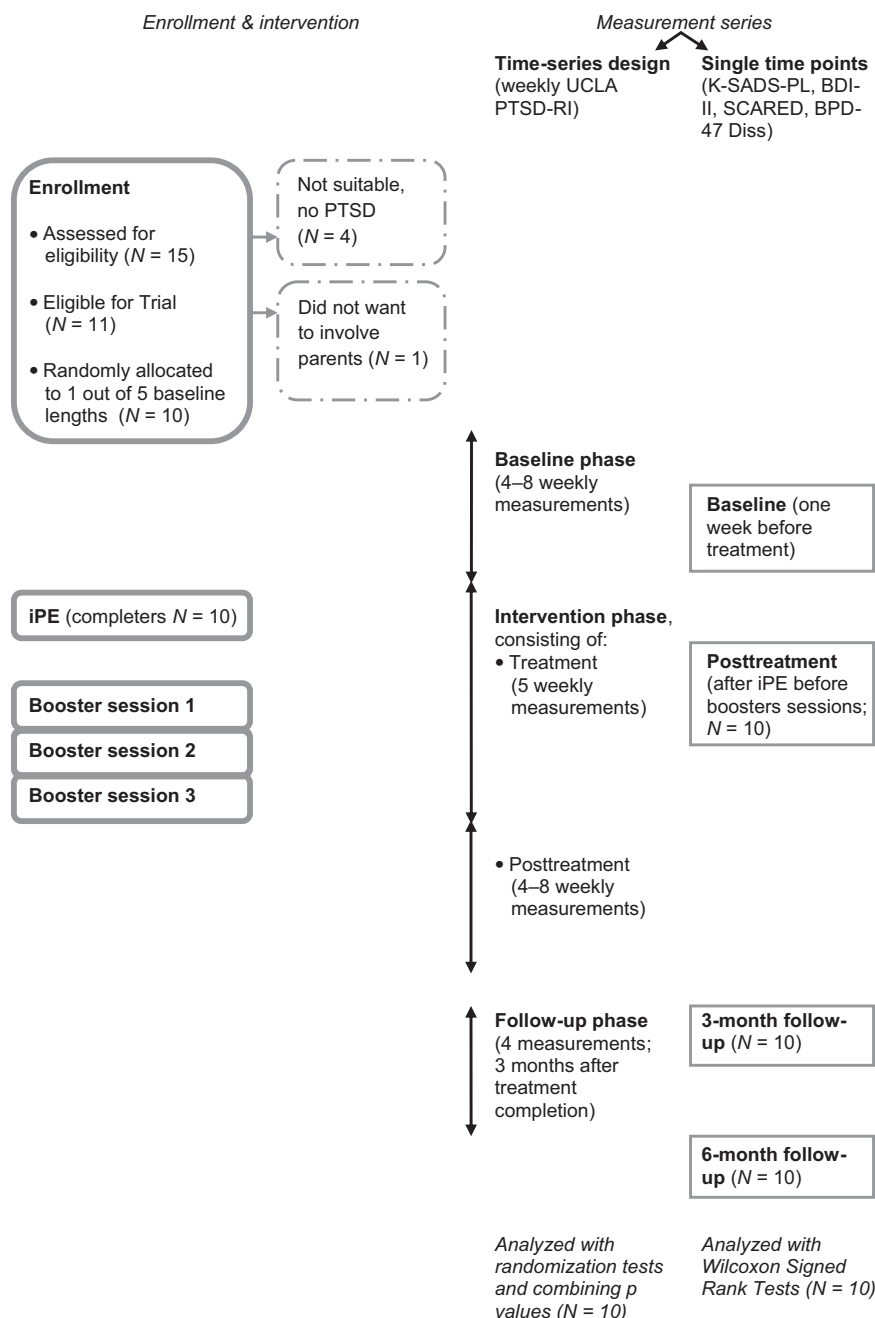


Figure 1 Flow diagram of participant recruitment and trial progress. BDI-II, Beck Depression Inventory Second Edition; BPD-47 Diss, Paranoid/dissociation subscale of the Borderline Personality Disorder symptom checklist; iPE, intensive Prolonged Exposure Therapy, first five treatment days in 1 week; K-SADS-PL, PTSD section (current episode) of the Kiddie-Schedule for Affective Disorders and Schizophrenia-Present and Lifetime Version; PTSD, Posttraumatic stress disorder; SCARED, Screen for Child Anxiety-Related Emotional Disorders; UCLA PTSD-RI, UCLA PTSD Reaction Index

the work of Thrasher, Power, Morant, Marks, and Dalglish (2010) that showed that social support in dealing with the effects of trauma is predictive of better treatment gains. Parents were trained to improve the quality of their social support when responding to the adolescent sharing the trauma narrative with the aim to reduce the adolescent's trauma-related avoidance behavior. The parent sessions were for the greater part conducted in the absence of the adolescents but at the end of the session, the adolescent was invited to also attend and share the traumatic narrative (s)he had processed that day with his/her parent(s) to provide them with the opportunity to apply their newly learned skills. One week after treatment completion, the adolescents received up to 3 weekly 90-min PE booster sessions within 1 month.

All participants were treated by a team of qualified therapists who were specially trained in PE; all had twice weekly group supervision from a senior therapist (AVM) for the duration of the trial.

Primary outcome measures

Self-reported PTSD severity over the last week was monitored using the University of California at Los Angeles PTSD Reaction Index (UCLA PTSD-RI; Steinberg, Brymer, Decker, & Pynoos, 2004), adolescent version. This 22-item self-report scale has been shown to have good to excellent internal consistency reliability (Cronbach's alpha between .88 and

.91) and convergent validity (Steinberg et al., 2013). Clinician-administered PTSD diagnostic status and symptom severity were measured by independent trained assessors, in accordance with the PTSD section (current episode) of the K-SADS-PL (Kaufman et al., 1997), whose interrater agreement is high, while its test-retest reliability κ coefficients represent good agreement (Kaufman et al., 1997).

Secondary outcome measures

Comorbid symptoms. Severity of depressive symptoms was assessed with the Beck Depression Inventory (BDI; Beck, Ward, Mendelson, Mock, & Erbaugh, 1961), a 21-item self-report questionnaire with high internal consistency (a mean Cronbach's alpha of .86; Beck, Steer, & Garbin, 1988). General anxiety symptoms were quantified using the 69-item self-report Screen for Child Anxiety-Related Emotional Disorders (SCARED), a measure showing good internal consistency (Cronbach's alpha between .74 and .93), test-retest reliability, and discriminative validity (Birmaher et al., 1997). Dissociative symptoms were evaluated with the eight-item 'Paranoid/Dissociation' subscale of the Borderline Personality Disorder symptom checklist (BPD-47 symptom checklist; Arntz & Dreesen, 1992), which self-report questionnaire (full version) has shown excellent internal consistency (Cronbach's alpha = .94; Arntz et al., 2003).

Dropout and safety. Dropout was recorded by the therapist and adverse events were monitored by self-report. Participants reported suicidal ideation, self-harm, and aggressive behavior on an 11-point Likert scale ranging from 'no, not at all' (0) to 'yes, very much' (10), while the use of alcohol and illicit or prescription drugs, crisis contacts with mental health providers, and psychiatric hospitalization were reported dichotomously (yes/no).

Design

As is shown in Figure 1, two different measurement series were performed: a time-series design that comprised frequent self-reports, and four single time points (baseline, posttreatment, 3-, and 6-month follow-up) that comprised a clinician-administered interview as well as self-report measures.

Time-series design. This time-series design is an ideal experimental strategy for evaluating a novel treatment approach, like our intensive PE, as it allows both the intervention process and outcome to be evaluated (Morgan & Morgan, 2001).

The severity of self-reported PTSD symptoms was assessed repeatedly: during the baseline phase and then continued during the intervention phase (weekly, 17 times in total). Baseline length varied between 4 and 8 weeks, with participants being randomly allocated to one of five baseline lengths (two participants per length). The intervention phase comprised a 5-week treatment phase (five consecutive weekdays of intensive PE, 1 week without treatment, and 3 weeks with weekly booster sessions) and a posttreatment phase that also varied from 4 to 8 weeks. Symptoms were further assessed (weekly, four measurements) during a 1-month follow-up phase, 3 months after treatment completion. Within this design, PTSD symptom severity scores during the intervention phase are compared with baseline scores.

Single time points. Clinician-administered PTSD diagnostic status and symptom severity, as well as self-reported comorbid symptoms, were evaluated at baseline (1 week before treatment), posttreatment (after intensive PE completion and before booster sessions), and at a 3-month and 6-month follow-up.

Data analysis

Time-series design. A randomization test was conducted for each participant. Randomization tests are nonparametric significance tests that are valid for single-case designs, without making distributional assumptions (Heyvaert & Onghena, 2014; Onghena & Edgington, 2005). The *a priori null hypothesis* for the randomization test was that intensive PE has no effect: participant responses are independent of the condition (i.e. 'baseline' vs. 'intensive PE') under which they are observed. The *alternative hypothesis* was that there are positive effects of the intensive PE on the UCLA PTSD-RI (directional, one-tailed), whereby a lower UCLA PTSD-RI scale score is expected in the intervention phase (B), compared to the baseline phase (A), for each participant. In accordance with this alternative hypothesis, the difference between the means for the baseline and intervention phase was applied as test statistic for the randomization test: $T = \bar{A} - \bar{B}$. The incorporation of randomization for each within-subject experimental design concerned the moment of phase change (i.e. the start of the intervention phase), resulting in five design possibilities: AAAAAABBBB, BBBBBBBB, AAAAAABBBBBBBBBB, AAAAAABBBBBBBB, and AAAAAABBBBBBBB.¹

The test statistic was calculated based on the data collected for each participant. For instance, for participant 1, the following UCLA PTSD-RI total scores were collected for the different time points in the A phase: 37; 31; 32; 30; and 36 and B phase: 17; 20; 15; 15; 16; 17; 17; 15; 16; 18; 20; and 15. Accordingly, the observed test statistic for participant 1 was $((37 + 31 + 32 + 30 + 36)/5) - ((17 + 20 + 15 + 15 + 16 + 17 + 17 + 15 + 16 + 18 + 20 + 15)/12) = 16.45$. Next, for each participant, the test statistic was calculated for all five design possibilities. These five test statistics were then sorted in ascending order, reflecting the randomization distribution under the null hypothesis. Following up on the example, for participant 1, the test statistics for the five possible permutations were 16.45, 13.77, 12.60, 10.69, and 9.14, respectively. The proportion of possible test statistic values that is as extreme as or even more extreme than the value of the test statistic based on the collected data is calculated and used to define the individual *p* value of the randomization test (Heyvaert & Onghena, 2014; Onghena & Edgington, 2005). The 10 individual *p* values that were the outcomes from the randomization tests were afterwards combined using the additive method (Edgington, 1972; Onghena & Edgington, 2005), and analyzed with the R software package developed by Bulté and Onghena (SCDA package; Bulté & Onghena, 2013).

Single time points. Analyses were conducted with statistical software (SPSS 22; IBM SPSS). Baseline scores on the measures assessing PTSD severity, depressive symptoms, general anxiety, and dissociative symptoms were compared with the posttreatment and 3- and 6-month follow-up scores using the nonparametric Wilcoxon signed rank tests because general assumptions for parametric testing were violated. Effect sizes were calculated with the formula $r = Z/\sqrt{N}$, where *N* is the number of observations, and indicated using Cohen's (1988) criteria (.1 = small effect; .3 = medium effect; and .5 = large effect).

Results

Sample characteristics

Our sample consisted of eight girls and two boys with a mean age of 15.90 years (*SD* = 1.52, *RANGE* 13–18). Table 1 shows their baseline demographic, trauma, and clinical characteristics. All 10 had experienced multiple interpersonal traumas in the

Table 1 Sample characteristics

Characteristics (<i>N</i> = 10)	
Education, No. (%)	
Lower level secondary school	5 (50)
Intermediate/higher level secondary school	4 (40)
Lower level tertiary education	1 (10)
Trauma history, No. (%)	
Multiple sexual abuse	7 (70)
Multiple violent (nonsexual) assault	9 (90)
Emotional abuse	10 (100)
Neglect	8 (80)
Time since trauma in years, mean (<i>SD</i>), range	9.30 (3.89), 1–13
UCLA PTSD-RI baseline, mean (<i>SD</i>), range	48.30 (6.50), 37–61
BPD-47 subscales baseline, mean (<i>SD</i>), range	
Affect dysregulation ^a	36.80 (10.56), 22–60
Negative self-concept ^b	21.30 (6.20), 12–29
Interpersonal problems ^c	8.30 (2.11), 6–13
Total amount of comorbid disorders, mean (<i>SD</i>), range	2.50 (1.27), 0–4
Axis I comorbidity (current), No. (%)	
1–2 comorbid disorder	9 (90)
≥3 comorbid disorders	3 (30)
Comorbid depressive disorder, No. (%)	6 (60)
Receiving psychotropic medication, No. (%)	7 (70)
Total treatment history ^d in months, mean (<i>SD</i>), range	6 (60)

BPD-47, Borderline Personality Disorder Checklist; UCLA PTSD-RI, UCLA PTSD Reaction Index.

^aSum score of the 'Mood', 'Impulsivity', and 'Anger' subscales.

^bAs measured by the 'Identity/Self-concept' subscale.

^cAs measured by the 'Relationships' subscale.

^dMental health care treatment history.

past: seven multiple sexual abuse (for six at age ≤ 12 years) and nine multiple violent (nonsexual) assaults (for eight at age ≤ 12 years). All reported affect dysregulation, negative self-concept, and interpersonal problems (BPD-47 subscales), resembling criteria for complex PTSD as proposed by the WHO ICD-11 (Cloitre, Garvert, Brewin, Bryant, & Maercker, 2013). Nine participants met criteria for a comorbid disorder: depressive disorder (*n* = 7) with some also reporting psychotic features (*n* = 2), other anxiety disorder(s) (*n* = 9), eating disorders (*n* = 3), conversion disorder (*n* = 1), autism spectrum disorder (*n* = 1), and alcohol dependence (*n* = 1). In addition, six participants reported suicidal ideations and four (urges of) self-harming behavior.

Primary outcomes

Self-reported PTSD severity. The individual graphs (Figure 2) show the UCLA PTSD-RI scores for all 10 participants during the baseline (A), intervention (B), and follow-up (C) phases.

Visual inspection suggests that PTSD symptoms decreased in the intervention phase and that UCLA PTSD-RI follow-up scores are lower compared to the

baseline values in all but one participant (participant 10). The results of the time-series analysis are displayed in Table S1, available online.

Applying Edgington's (1972) additive method, the combined *p* value across all 10 participants showed a statistically significant decrease of PTSD symptoms (*p* = .002). Together, our results show that PTSD symptoms were significantly lowered by the intensive PE.

PTSD diagnostic status and symptom severity (clinician-administered interview). Table 2 lists the descriptive statistics. There was a significant baseline-to-posttreatment reduction in K-SADS-PL-PTSD section total scores (*z* = −2.81, *p* = .005) with a large effect size (*r* = .63), as well as for baseline-to-3-month follow-up (*z* = −2.81, *p* = .005, *r* = .63) and 6-month follow-up (*z* = −2.81, *p* = .005, *r* = .63). The number of adolescents meeting the K-SADS-PL-PTSD diagnosis decreased from 10 to 6 (40% remission) from baseline to posttreatment and from 10 to 2 (80% remission) from baseline to follow-up at 3 and 6 months.

Secondary outcomes

Comorbid symptoms. Table 2 lists the descriptive statistics. There was a significant reduction in depressive symptoms (BDI-II total scores) from baseline to posttreatment (*z* = −2.24, *p* = .025) with a large effect size (*r* = .50) as well as to the 3-month (*z* = −2.65, *p* = .008, *r* = .59) and 6-month follow-ups (*z* = −2.81, *p* = .005, *r* = .63). Furthermore, there was a significant baseline-to-posttreatment reduction in general anxiety symptoms (SCARED total scores) (*z* = −2.40, *p* = .017) with a large effect size (*r* = .54), as well for the baseline-to-3-month (*z* = −2.70, *p* = .007, *r* = .60) and 6-month follow-ups (*z* = −2.50, *p* = .013, *r* = .56). We also computed a significant reduction in dissociative symptoms (BPD-47, paranoid/dissociation subsection score) from baseline to posttreatment (*z* = −2.81, *p* = .005) with a large effect size (*r* = .63), as well as to the 3-month (*z* = −2.60, *p* = .009, *r* = .58) and 6-month follow-ups (*z* = −2.61, *p* = .009, *r* = .58).

Dropout and safety. None of the participants dropped out of the treatment and no adversities (suicidal behavior, crisis contact, or psychiatric hospitalization) occurred during the intervention phase. In addition, none of the participants showed increased PTSD symptoms (see Figure 2 for the individual graphs). There were (nonsignificant) reductions in suicidal ideation (baseline *M* = 3.00, post *M* = 2.20), self-harm (baseline *M* = 2.20, post *M* = 1.00), and aggressive behavior (baseline *M* = 0.20, post *M* = 0.00), and none of the participants reported having abused alcohol, drugs, or prescription medication during the intervention phase.

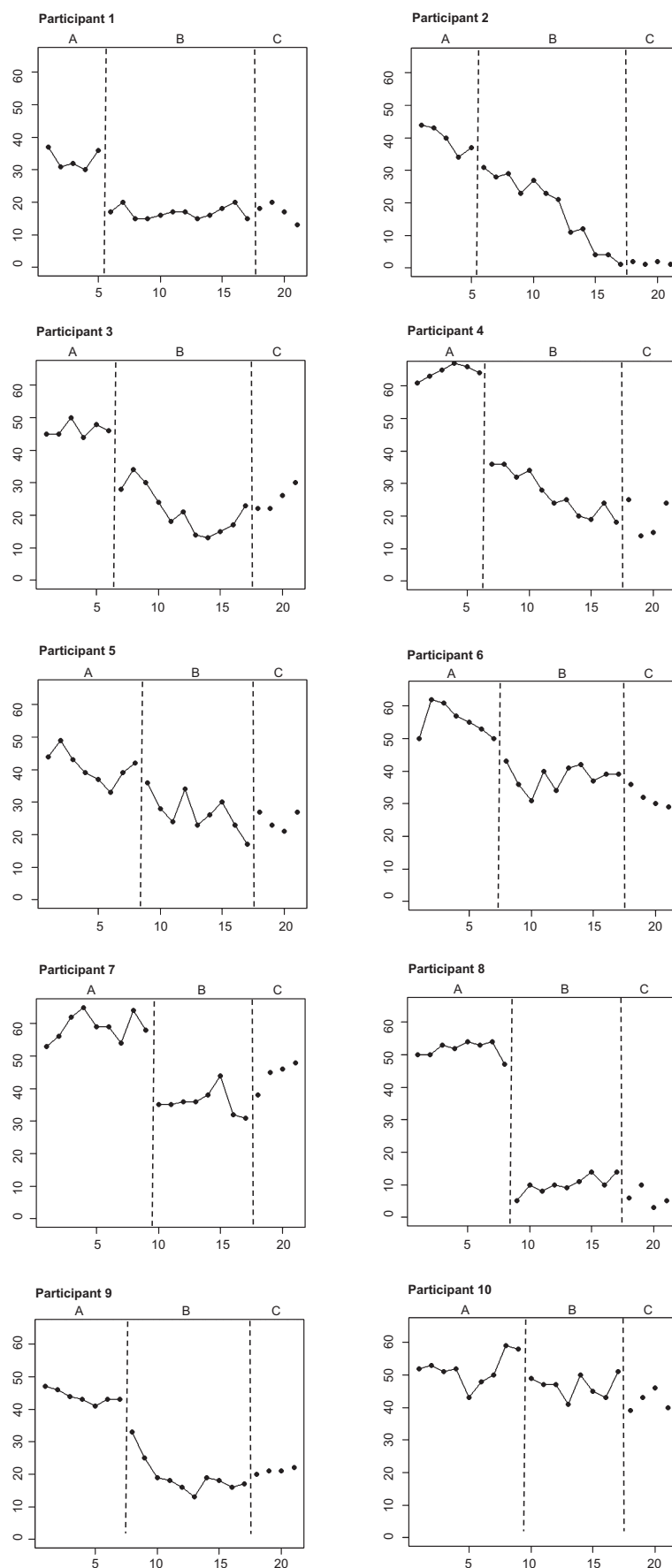


Figure 2 Primary outcome variable: weekly measures of PTSD symptom severity (total UCLA PTSD-RI scores) across the baseline (A), intervention (B), and follow-up (C) phases. x-axis = measurement occasions; y-axis = outcome scores; phases are divided by vertical lines

Table 2 Means, medians, and SDs for Wilcoxon signed rank tests for the primary and secondary outcomes ($N = 10$)

	Baseline			Posttreatment			3-month follow-up			6-month follow-up		
	Mean	Median	SD	Mean	Median	SD	Mean	Median	SD	Mean	Median	SD
K-SADS-PL	31.60	32.00	3.10	26.40*	26.00	4.12	23.20*	23.00	3.55	23.30*	22.00	4.99
BDI-II	32.50	35.50	14.97	21.70*	20.50	14.94	20.10*	21.00	13.76	17.00*	12.00	13.40
SCARED	71.40	79.00	27.99	45.20*	51.50	22.39	37.70*	44.00	24.87	41.50*	45.50	25.27
BPD-47 Diss	21.90	22.00	7.06	14.30*	12.50	5.54	13.10*	11.00	6.45	12.50*	11.00	5.93

BDI-II, Beck Depression Inventory Second Edition; BPD-47 Diss, Paranoid/dissociation subscale of the Borderline Personality Disorder symptom checklist; K-SADS-PL, PTSD section (current episode) of the Kiddie-Schedule for Affective Disorders and Schizophrenia-Present and Lifetime Version; SCARED, Screen for Child Anxiety-Related Emotional Disorders.

* $p \leq .05$ compared to baseline.

Discussion

Our results of this first proof of concept trial investigating an intensive PE therapy program targeting adolescents (aged 12–18 years, $N = 10$) with complex PTSD and comorbid disorders, suggest that the intervention was effective in reducing PTSD symptoms and achieving remission of PTSD diagnosis. In addition, in line with the results of the open studies into intensive TFT (Matulis et al., 2014; Silverstone et al., 2016), symptoms declined within a short period of time, with our study showing gains persisting over time (6 months) as well. Our results were also in agreement with studies showing the effectiveness of weekly-based PE in adolescents with PTSD after single trauma (Gilboa-Schechtman et al., 2010) and expand the evidence base on the effectiveness of PE for adolescents having suffered multiple interpersonal traumas before the age of 12 and who show more complex pathology indicative of complex PTSD as well as high levels of comorbid disorders (Foa et al., 2013). In addition, along with the PTSD symptom reduction, also symptoms of depression, general anxiety, and dissociation were reduced significantly, even though these were not directly addressed in the treatment. The amelioration of comorbid depressive symptoms we observed is consistent with the findings of a meta-analysis evaluating the effectiveness of TFT on symptoms of PTSD and comorbid depression in children and adolescents (Lenz & Hollenbaugh, 2015), which showed TFT to be more effective in reducing depressive symptoms than other (nontrauma-focused) treatments. Lastly, our data are also in line with studies in adult PTSD populations that consistently found that secondary symptoms decreased along with the PTSD symptoms with exposure-based treatments (Van Minnen, Zoellner, Harned, & Mills, 2015).

One of the reasons that prompted us to develop a short and intensive PE program for adolescents was to prevent dropout. The fact that, despite their complex PTSD and comorbid problems, all 10 participants completed the program shows that intensive PE does just that, whereas dropout rates in other studies treating children and adolescents with PTSD for longer periods were relatively high (Cohen

et al., 2011; Gilboa-Schechtman et al., 2010). Moreover, given that, on average, 18% of adult PTSD patients drops out of treatment (Imel et al., 2013), with some studies reporting rates as high as 52%, and that dropout rates range from 28% up to 75% for pediatric outpatient mental health care in general (De Haan, Boon, de Jong, Hoeve, & Vermeiren, 2013), our results, if corroborated, have important clinical implications. Even though some clinicians are concerned that with its quick succession of trauma-focused interventions, intensive PE treatment may worsen symptoms or induce adverse events (Tarrier et al., 1999), the results of our trial show that, despite the massed sessions, our intensive PE protocol was safe and did not cause any adverse events.

In addition, by achieving significant symptom reduction in adolescents in just 1 week, intensive PE, if implemented soon after the trauma, may reduce the likelihood of developing the severe long-term consequences typically associated with PTSD (Kessler, 2000), such as the increased risk of revictimization (Risser, Hetzel-Riggin, Thomsen, & McCanne, 2006) or suicidal attempts (Wilcox, Storr, & Breslau, 2009), while, ideally, further benefits of rapid symptom reduction, such as a timely return to school and prevention of developmental stagnation, may be anticipated.

As to limitations, obviously the small sample size must be mentioned. The small sample size does restrict the generalizability of our findings (Gliner, Morgan, & Harmon, 2000), although the sample is highly representative of adolescents with complex PTSD, as typically seen in clinical practice. Furthermore, it was not possible to replicate the intervention by offering the treatment again separated by an adequate washout period, as is typically done in experimental studies (Onghena & Edgington, 2005), which lowers the external validity of our findings as well. Another limitation is that no control or placebo treatment was offered to indicate that the observed treatment effects were treatment specific instead of ‘attention alone’. Note, however, that all adolescents already received mental health care treatment (with an average of 2.5 years of treatment) when entering the intensive PE, which

suggests that it is unlikely that the findings should be attributed to placebo or nonspecific effects of treatment (rather than exposure effects). We further point out that, although we did not offer additional nontrauma-focused modules to the adolescents, the intensive PE did include parent sessions, the effects of which we have not elucidated. Dismantling studies will need to establish whether parent sessions are critical to intensive PE outcome. Another limitation is that the chosen design with weekly repeated assessments is vulnerable to carryover effects; however, visual inspection of the results provides no clear indication for those test effects in the current study.

The treatment outcomes we obtained and the absence of dropout and adverse events may have important implications for clinical practice as together these results expand the effectiveness of TFT – or more specifically PE – for young PTSD patients suffering from more complex pathology as a result of multiple interpersonal trauma. Importantly, our intervention did not include any nontrauma-focused techniques, showing that, despite hesitations of clinicians (Allen & Johnson, 2011; Van Minnen et al., 2010), intensive TFT can be effectively and safely applied in adolescent populations with complex PTSD and comorbid mental problems. Our work thus contrasts with the recommendation to start treatment with a stabilizing phase before initiating TFT especially when traumas were repeated and interpersonal or when patients are suffering

from complex PTSD psychopathology (Leenarts et al., 2013).

Although the cohort size is small, with our results we are, to our knowledge, the first to demonstrate the effectiveness and safety of a stand-alone intensive PE treatment targeting a complex pediatric PTSD population and with the favorable results of this first proof of concept trial offering strong support for a randomized controlled study.

Supporting information

Additional Supporting Information may be found in the online version of this article:

Table S1. Results of the time-series analysis.

Acknowledgements

This work was supported by ZonMw, the Netherlands Organization for Health Research and Development. The authors have declared they have no competing or potential conflicts of interest to declare. They thank all participants, therapists, research assistants, and all others who contributed to this study.

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Key points

- Current treatment for PTSD, including prolonged exposure (PE), usually lasts several months with sessions being delivered on a weekly basis.
- In this study, PE is administered in an accelerated way for adolescents with complex PTSD.
- Intensive PE reduced PTSD symptoms in just 1 week.
- There was neither treatment dropout nor any symptom exacerbation or adverse events.
- The results of this first proof of concept trial suggest that intensive PE is effective, safe, and well tolerated.

Note

1. Note that for the first design possibility, the A phase consists of five measurement occasions although the baseline for the first permutation consisted of 4 weeks. The measurement occasion on the first day of the fifth week was also labeled 'A' because the UCLA PTSD-RI, completed the first day of each week, assessed PTSD symptoms within the last week and an intervention effect during the actual intensive PE phase (first week) was not expected.

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Accepted for publication: 5 April 2017

First published online: 14 June 2017